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SEPM - DevOps Assignment (Case Study)

At its core, DevOps is the combination of cultural philosophies, practices & tools designed to increase an organization's ability to deliver applications & services at high velocity. By bringing development & operations teams together, organizations can evolve & improve products at a much faster pace than traditional software development models allow.

I] The CALMS Framework:

- Culture:

The foundation of DevOps is collaboration - sharing a common goal, swarming on problems together, & treating product stability & development as everyone's job, not just one team's responsibility.

- Automation:
- By eliminating repetitive manual work, teams create reliable systems that catch bugs an security flaws much sooner in the lifecycle.

- Lean:

A DevOps mindset embraces failure & continuously looks for opportunities to improve. Teams assume things will occasionally go wrong, so they build systems for fast detection & rapid recovery.

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- Measurement:
Teams track critical data to make decisions such as how long it takes to move from development to deployment, how often failures occur, & how long it takes to recover.

- Sharing:
Successes & rough patches are shared equally. Pairing developers & operators throughout the application's lifecycle builds trust & enables better knowledge transfer.

II) DevOps Practices:
The technical execution of DevOps relies on a heavily automated pipeline where the order of operations is critical.

i) Continuous Development - Planning & Coding
- This initial phase involves planning the software's functionality, writing the code, & utilizing version control systems to manage the codebase.

ii) Continuous Integration - Automated Builds
- Everytime a developer commits a new revision to the codebase, it automatically triggers a build & unit test sequence. This ensures that bugs are found & addressed much quicker.

iii) Continuous testing - Quality assurance
- Automated tests are continuously & repeatedly executed against the codebase across various deployment environments to ensure quality & improvement.

iv) Continuous Delivery & Deployment
- Because code is rapidly built & tested via automation, known "good" code becomes part of the software product. In continuous delivery, the final push to production is manual, whereas continuous deployment automates this final step entirely.

v) Continuous Monitoring
- Monitoring technologies are incorporated to continuously ensure the reliability, performance & health of the application & infrastructure in the production environment.

Architectural Support:
To make this pipeline run smoothly, DevOps relies heavily on Infrastructure as Code (IaC) managing server configurations via text files rather than manual setup, & microservices, which break monolithic applications into smaller, easily deployable components.

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III) DevOps Engineer: Role & Responsibilities

A DevOps Engineer acts as the bridge between software creation & system stability. They are tasked with ensuring the pipeline above runs efficiently & securely.

Key Responsibilities:

- Managing & automating the CI/CD pipelines
- Managing infrastructure & assessing performance
- Scripting, coding & writing specifications & documentation.
- Monitoring system health.

Required skills & experience:

- Technical knowledge: Proficiency in open-source automation tools (like Git, Docker & Kubernetes), cloud services (AWS, Azure, Google Cloud), & databases (SQL or NoSQL).
- Coding & Scripting: Java, C++, Python; Bash, Powershell
- Background: BS/MS in CS, 2+ years of experience as a developer/system admin, strong data management capabilities.
- Personal Attributes:

Beyond technical skills, succesfull DevOps engineers need strong communication & leadership abilities, highly informed analytical skills, empathy for other team members & the ability to thrive in cross-functional teams.